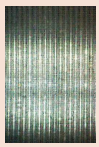
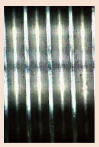
Spindle Pitch



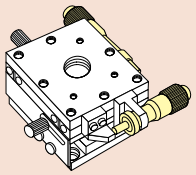
Pitch=0.1 mm



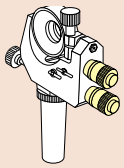
Pitch=0.5 mm

Typical Applications

- Semiconductor-wafer positioning machinery and optical component alignment units, etc.
- Precision X-Y table positioning



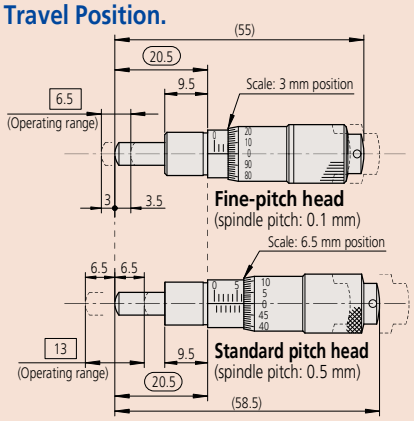
- Precision adjustment of mirror in holder



Precision adjustment of mirror in holder



Comparison of Mounting Dimensions Between a Fine-pitch Head and a Standard-pitch Head at the Mid-range Travel Position.



Note: While the fine-pitch micrometer head has a measuring range of 6.5 mm, the standard head has a larger range of 13 mm.
When replacing a standard head, the fine-pitch type can use the common range in the middle of the spindle travel. The standard and compact types of fine-pitch head are otherwise completely interchangeable.

Micrometer Heads
SERIES 148 — Fine Spindle Feed of 0.1 mm/rev

- Fine spindle feeding of just 0.1 mm/rev (one fifth compared with standard model). Suitable for extra-fine adjustment and positioning.
- External dimensions are compatible with standard 0.5 mm pitch heads.
- Suitable for the fine feeding of precision stages on semiconductor equipment and optical-axis alignment device.

SPECIFICATIONS

Metric

Code No.	Range (mm)	Graduation (mm)	Stem dia. (mm)	Stem	Spindle end	Spindle pitch (mm)	Maximum permissible error J_{MPE} (µm)	Special features
148-142-10	0 - 6.5	0.002	9.5	Plain	Spherical (SR4)	0.1	±2	—
148-143-10				W/clamp nut				
148-342-10				Plain				Thicker & shorter thimble
148-343-10				W/clamp nut				
148-242-10			6	Plain	Spherical (SR3)		±5	Small thimble diameter
148-243-10				W/clamp nut				
148-244	0 - 5	0.004	3.5	Plain	Spherical (SR1.5)			
148-245				W/clamp nut				

- Measuring face: Material/Alloy tool steel, Hardness/60 HRC or more, Lapped
- Scale finishing: Satin-chrome plated

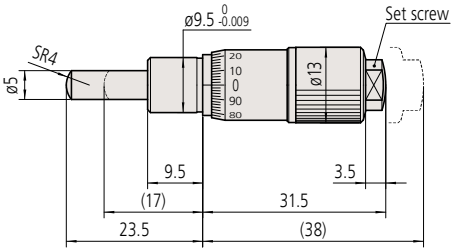
DIMENSIONS

Plain Stem

Unit: mm



Plain stem Spherical face





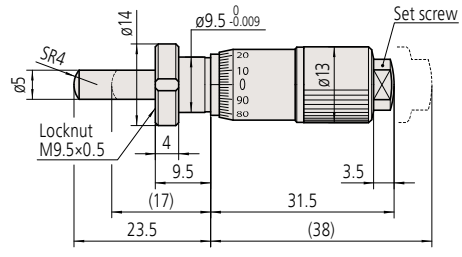
Sleeve marker

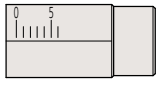
148-142-10 Mass: 31 g

Stem Locknut



Stem locknut Spherical face





Sleeve marker

- Fixture thickness: 6 mm

148-143-10 Mass: 34 g

( ): with spindle fully retracted.

Mitutoyo reserves the right to change any or all aspects of any product specification, including prices, designs and service content, without notice.

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Micrometer Heads
SERIES 148 — Fine Spindle Feed of 0.1 mm/rev

DIMENSIONS

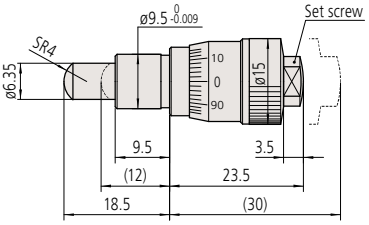
Unit: mm

Plain Stem




Plain stem

Spherical face



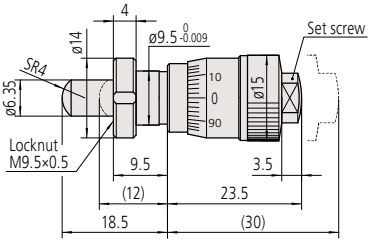
148-342-10 Mass: 29 g

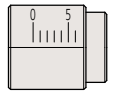
Stem Locknut




Stem locknut

Spherical face




Sleeve marker

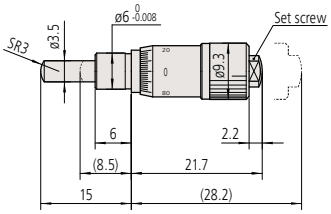
• Fixture thickness: 6 mm
148-343-10 Mass: 31 g

Plain Stem




Plain stem

Spherical face



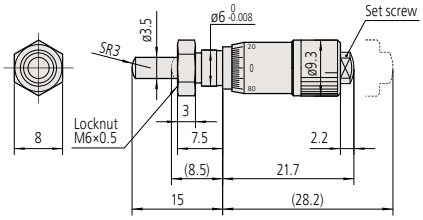
148-242-10 Mass: 10 g

Stem Locknut




Stem locknut

Spherical face




Sleeve marker

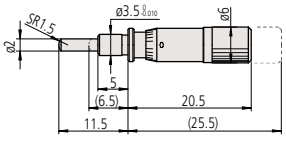
• Fixture thickness: 4 mm
148-243-10 Mass: 10 g

Plain Stem




Plain stem

Spherical face



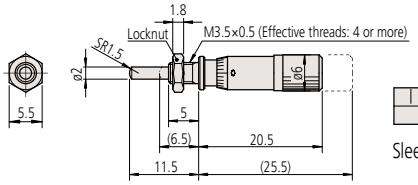
148-244 Mass: 4 g

Stem Locknut




Stem locknut

Spherical face




Sleeve marker

• Fixture thickness: 3 mm
148-245 Mass: 5 g

( ): with spindle fully retracted.

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